# In-building cellular coverage test: Example: carrier cellular DAS, warehouse (completed example)

Example distribution warehouse, Arlington VA area (fictional) · Example County Building Official (example) · Owner spec: carrier coverage, Common preset · prepared 2026-09-30

## Result: PASS for AT&T, Verizon and T-Mobile: every square at the cellular target (Commercial cellular coverage, below)

### Building and people

| | |
|---|---|
| Building | Example: carrier cellular DAS, warehouse (completed example) · Example distribution warehouse, Arlington VA area (fictional) · 1 floors · 120,000 sq ft · Traditional build, worst 5% (28 dB) |
| AHJ / fire marshal | Alicia Grant · Example County Building Official (example) · 703-555-0146 · agrant@buildingofficial.example |
| Building owner | Ray Kim · Sample Logistics LLC (example) · 703-555-0151 · rkim@samplelogistics.example |
| Our technician | Jordan Lee · Sample RF Services (example) · 703-555-0109 · jlee@samplerf.example |

### Tester and instruments

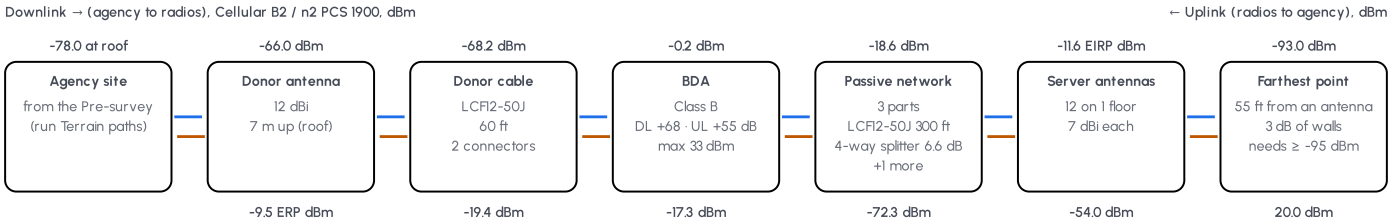
| | |
|---|---|
| Tester | Jordan Lee (example) |
| FCC GROL | PG-EX-0001 (example) |
| NICET | Level III (example) |
| Test dates | 2026-09-22 to 2026-09-23 |

| Instrument | Serial | Calibrated | Status |
|---|---|---|---|
| Nextivity COMPASS XR | EX-2001 | 2026-05-01 | current |

### Compliance

| | |
|---|---|
| Licensee / agency consent to retransmit (47 CFR 90.219(b)(1)) | 2026-08-11 |
| AHJ permit | 2026-07-01 |
| Design approved | 2026-07-24 |
| Acceptance test passed | 2026-09-30 |
| Annual retest due | 2027-09-30 |
| Compared with | the acceptance test of 2026-09-30 |
| Material rules confirmed | 22: UL 2524 listing of active equipment; Enclosures for BDAs and batteries; Secondary (standby) power; Supervisory signals to the fire alarm system; Dedicated annunciator; Backbone survivability (fire resistance); Older-edition survivability rules (jurisdictions still on them); Cable fire ratings (plenum / riser); Mechanical protection of coax; Lightning / surge protection on the donor line; Grounding the coax shield at entry; Donor antenna mounting and isolation; Oscillation control; FCC Part 90 signal booster class, consent and registration; Frequencies and upgradeability; Near-far and antenna density; Link budget: choosing cable size; Component power ratings; Low-PIM choices for multi-carrier DAS; Radiating cable stop bands; Fiber connectors for RF-over-fiber DAS; Labelling and signage |

### System diagram: signal path, start to finish




Levels are the Cellular B2 / n2 PCS 1900 link budget's worst case (1990 MHz down, 1910 MHz up); every band is in the table below. Blue: what reaches the radios inside; orange: what reaches the agency's receiver.

| Band | Systems | At roof dBm | BDA out dBm | Antenna EIRP dBm | Farthest point dBm | Needs dBm | Uplink at site dBm | Checks |
|---|---|---|---|---|---|---|---|---|
| Cellular B2 / n2 PCS 1900 | Verizon, AT&T, T-Mobile | -78.0 | -0.2 | -11.6 | -93.0 | -95.0 | – | pass, 1 warn |
| Cellular B13 / n13 Upper 700 | Verizon | -88.0 | -7.3 | -16.8 | -89.7 | -95.0 | – | pass, 1 warn |
| Cellular B71 / n71 600 MHz | T-Mobile | -84.0 | -9.3 | -19.2 | -90.9 | -95.0 | – | pass, 1 warn |

No donor paths yet: run Terrain paths on the Pre-survey to add each agency site's distance, bearing and terrain.

**Antennas by floor:** Floor 1: 12

## Commercial cellular coverage

Reported apart from the emergency responder result above and never counted in it. Target: Common: RSRP ≥ -95 dBm (≥ -100 dBm from 3.3 GHz), SINR ≥ 5 dB on 95% of the tested squares, carriers: AT&T, Verizon, T-Mobile. Owner specs and RFPs cluster at RSRP –95 dBm over 95% for LTE and –100 to –105 dBm for C-band; SINR runs 5–10 dB in US documents. Guidance, not a carrier spec. Each square counts the carrier's best cell.

| Floor | Carrier | Layer | Tested | Pass | % | Result |
|---|---|---|---|---|---|---|
| 1 | AT&T | below 3.3 GHz | 20 | 20 | 100% | pass |
| 1 | T-Mobile | below 3.3 GHz | 20 | 20 | 100% | pass |
| 1 | Verizon | below 3.3 GHz | 20 | 20 | 100% | pass |

## Rooftop donor

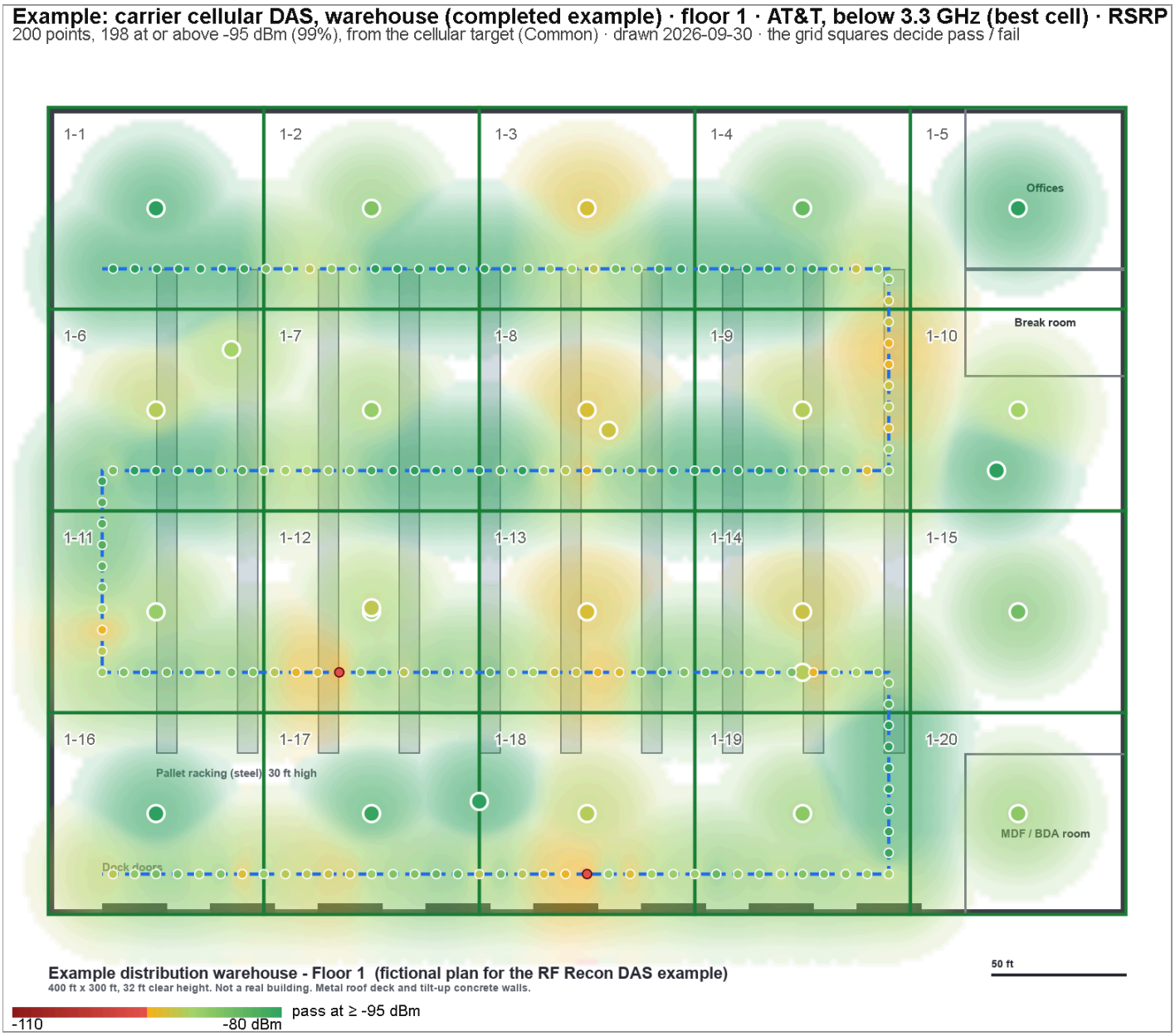
| Carrier | RSRP dBm | SINR dB | Band | Azimuth | Height ft | Measured | Against Providence §3.9 |
|---|---|---|---|---|---|---|---|
| AT&T | -78 | 11 | 2 | 215° | 38 | 2026-06-16 | meets –85 dBm target |
| Verizon | -88 | 6 | 13 | 160° | 38 | 2026-06-16 | minimum only (–95 dBm) |
| T-Mobile | -84 | 7 | 71 | 20° | 38 | 2026-06-16 | meets –85 dBm target |

## Carrier approvals

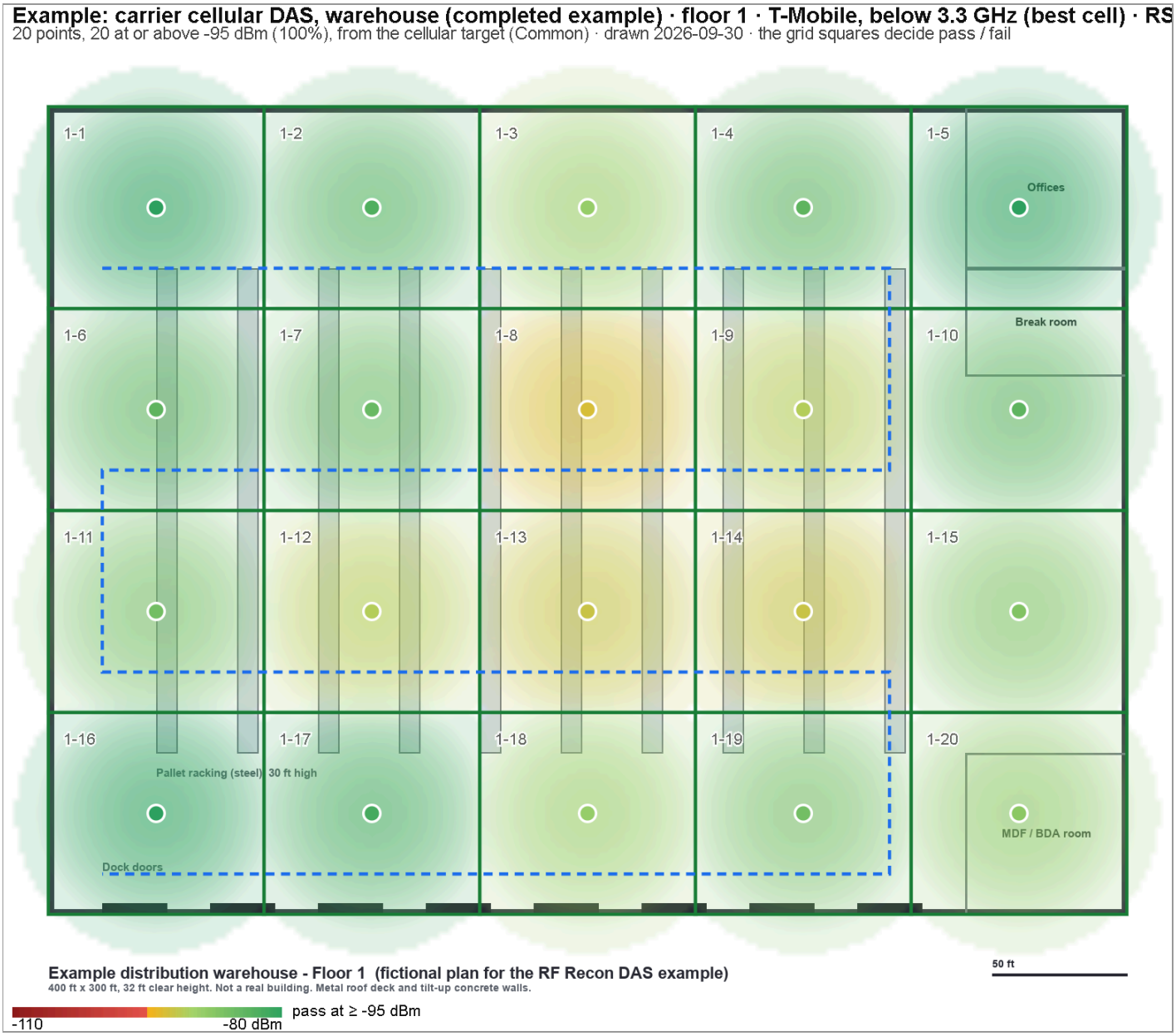
| Carrier | Path | Asked | Consent | Contact / reference |
|---|---|---|---|---|
| AT&T | Industrial booster or off-air DAS | 2026-06-22 | 2026-07-20 | AT&T DAS team (example), ref ATT-EX-0442: B2 consented |
| Verizon | Industrial booster or off-air DAS | 2026-06-22 | 2026-07-28 | Verizon account manager (example): B13 consented |
| T-Mobile | Industrial booster or off-air DAS | 2026-06-23 | 2026-08-11 | Rebroadcast Agreement signed (example), ref TMO-EX-1187: B71 |

## Walk test heat maps

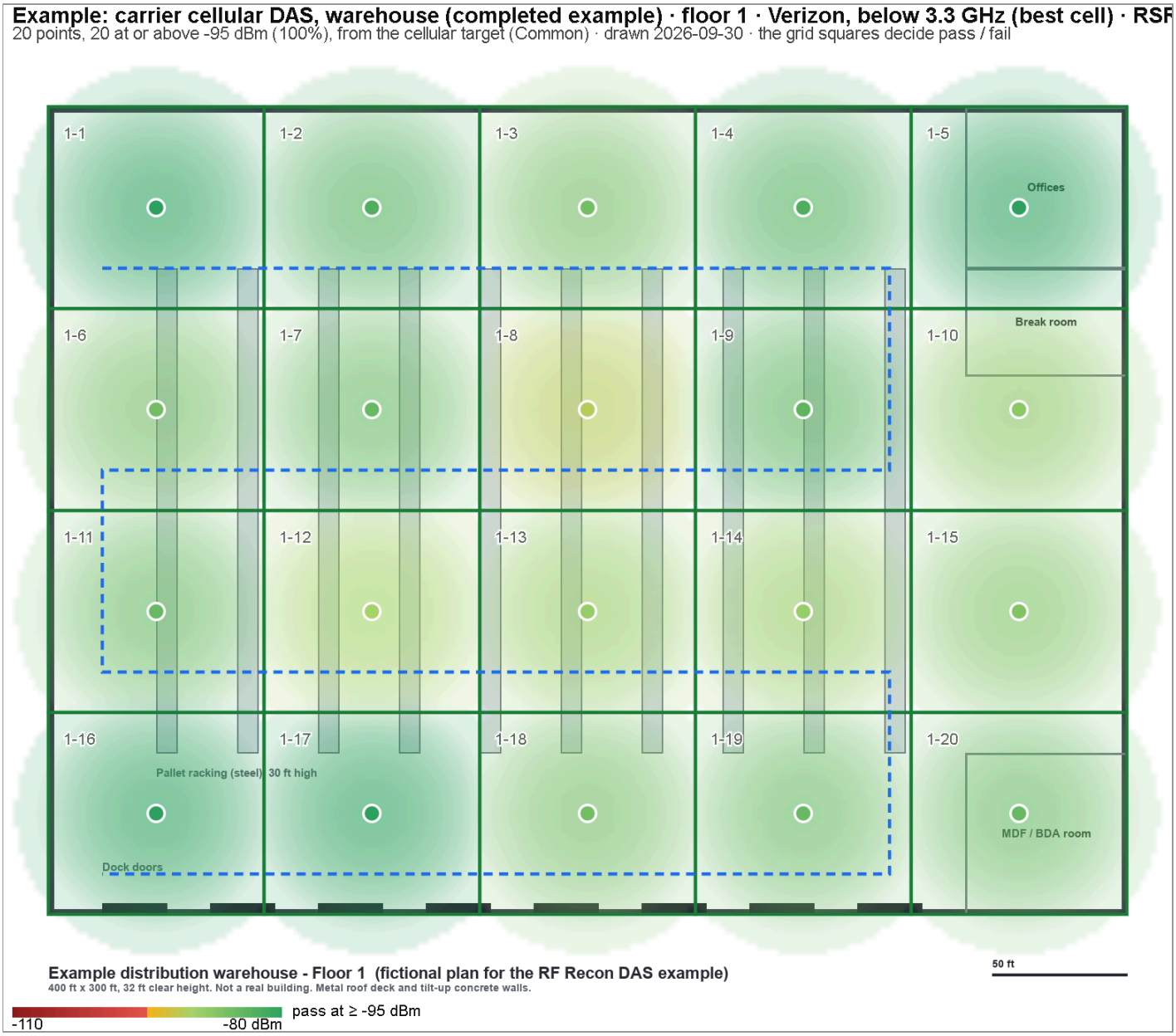
Readings walked on each floor's plan, coloured against the requirement. The heat map shows every reading, calibrated or not; the grid squares above decide pass / fail.




Floor 1 · AT&T, below 3.3 GHz (best cell) · RSRP dBm · 200 points, 198 at or above -95 dBm (99%), from the cellular target (Common) · drawn 2026-09-30 · the grid squares decide pass / fail




Floor 1 · T-Mobile, below 3.3 GHz (best cell) · RSRP dBm · 20 points, 20 at or above -95 dBm (100%), from the cellular target (Common) · drawn 2026-09-30 · the grid squares decide pass / fail




Floor 1 · Verizon, below 3.3 GHz (best cell) · RSRP dBm · 20 points, 20 at or above -95 dBm (100%), from the cellular target (Common) · drawn 2026-09-30 · the grid squares decide pass / fail

This product uses the FCC Data API but is not endorsed or certified by the FCC.